

while Maven Systems specialises in automated lighting and smart meters. There are some start-ups with even bigger ideas. One among them is Quenext, which aims to consolidate and analyse data from innumerable systems ranging from weather to power, to help utilities optimise their supply and reduce losses. LoudCell develops IoT-based sensors, remote hardware and cloud-based middleware, which are coupled with an intuitive dashboard to provide customers with analytics and intelligence to understand and reduce their power consumption.

A low-cost smart irrigation controller from FlyBird Farm Innovations helps farmers to irrigate their crops precisely, depending on weather, soil conditions and crop requirements. One more interesting solution for this sector comes from TartanSense, which aims to use unmanned aerial vehicles and machine learning to bring actionable intelligence to the agricultural sector.

Zippr is a Telangana-based company that creates eight-digit alphanumeric codes to replace traditional door numbers. These unique IDs are mapped to precise locations overlaid on a digital map. This fundamentally transforms how locations are referenced, shared and navigated. Zippr won the IBM Smart Camp for Smart City 2016, and is working with the Telangana government to implement this smart address system in Hyderabad.

Taxi services like Ola and Uber are playing their part in building smart cities by providing innovative ride options, including sharing and shuttle services. These bring down the cost of rides, improve commute facilities and reduce pollution through car-pooling.

Programmes such as the AIM Smart City Accelerator provide mentorship, guidance and planning support for start-ups and other agencies working towards the smart city dream!

Opportunity is knocking at our doors

McKinsey Research projects the smart city industry to be a \$400 billion market by 2020, with 600 cities worldwide, which are together expected to

generate 60 per cent of the world's GDP by 2025.

A Forbes article by Mohit Kochar of KPIT Technologies explains how smart cities are India's next technology opportunity to lead the world. He states rapid urbanisation, excellent ecosystem, manpower and tech prowess, the third largest start-up base in the world and a reputed brand image in information technology as reasons why we are well poised to capitalise this opportunity.

Clearly, there is no dearth of technologies required for smart cities in India too. However, we need to focus on proper planning and infrastructure development. The investment also needs to be thoughtful and transparent. It is estimated that the first 100 smart cities in India will require an annual investment of ₹ 350 billion over the next 20 years! This can simply not be achieved without private investments and public-private partnerships. There must also be excellent collaboration between central, state and local authorities.

Private and government agencies must also understand that such cities cannot be built without the participation of people. This is being reiterated time and again by experts in the field. People come in at the very beginning of the process—because solutions for a smart city have to be built around what people need and not merely what other cities have done! The residents of a city have to be involved all through the processes of conceptualisation, planning, development and implementation.

The smart city infrastructure must also be easy for people to use. For example, we need to remember that connectivity is poor, so implementers have to think about how to work around it. Literacy levels being low, you cannot expect everybody to type in passwords or fill in online forms. Methods like facial recognition or biometrics have to be used to authenticate individuals and validate data. In such and many other ways, government officials have to ensure that people can transition smoothly to this connected world—because, whether it is in 2025 or later, cities are inevitably going to get smarter! **EPF**



LOW COST LCD / LED MODULES



LC035



LC135

Specifications

3½ Digits, 1999 Counts (max.) LCD	3½ Digits, 1999 Counts (max.) LED
LC035	LC135
0 - 200mV DC 0 - 2, 20, 200V DC 0 - 200µA DC	0 - 2, 20, 200mA DC
Auxiliary Power Supply	
9VDC ± 10%	5VDC ± 10%

Applications

• Ammeter	• Watt Meter
• Voltmeter	• LUX Meter
• Thermometer	• LCR Meter
• Power Supply	• Capacitance Meter
• PH Meter	• Other Industrial and Domestic uses.
• DB Meter	

An ISO 9001:2008 Certified Company
MECO INSTRUMENTS PRIVATE LTD.
 Plot No. EL-1, MIDC Electronic Zone,
 TTC Industrial Area, Mahape,
 Navi Mumbai - 400710 (INDIA)
 Tel : 091-22-2767 3300 (Board), 2767 3311-16 (Sales)
 Fax : 091-22-2767 3310, 2767 3330
 Email : sales@meconinst.com
 Web : www.meconinst.com